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IT4010 Research Project – 2026 Jan

Topic Assessment Form

V2.0

Project ID: **R26-IT-005**

1. Topic (12 words max)

Water Purification

2. Research group the project belongs to

AIMS - Autonomous Intelligent Machines and Systems

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (500 words max) – references not included in word count.

Access to safe, clean, and health-appropriate drinking water remains a major global challenge despite significant advancements in water purification technologies. Rapid urbanization, industrial discharge, agricultural runoff, climate change, and poor water resource management have led to widespread contamination of surface and groundwater sources with heavy metals, fluoride, agrochemicals, dissolved salts, and microbial pathogens. According to the World Health Organization (WHO), contaminated drinking water is responsible for numerous waterborne diseases and long-term health complications worldwide.[1][3]

Conventional water purification methods such as boiling, chlorination, Reverse Osmosis (RO), activated carbon filtration, and distillation are commonly used to address water quality issues. However, these systems present several critical limitations. RO systems are associated with high water wastage and removal of essential minerals, distillation is energy-intensive, and chemical-based treatments may produce harmful by-products. Moreover, most existing systems lack real-time water quality monitoring, intelligent decision-making, and adaptive control mechanisms, resulting in inefficient operation and limited user awareness.[2]

In Sri Lanka, the problem is particularly severe. A significant portion of the population depends on groundwater, wells, and municipal water supplies that are often contaminated with excessive fluoride, heavy metals, and agricultural chemicals. The prevalence of chronic kidney disease of Unknown Origin (CKDu) in the North Central

Province has been strongly associated with prolonged consumption of poor-quality drinking water. Additionally, frequent floods and aging water distribution infrastructure increase the risk of microbial contamination, further threatening public health. While RO-based household purification systems are increasingly adopted, they operate without intelligent verification of water quality, mineral balance, or energy efficiency.[4][5]

The core research problem addressed in this study is the lack of an integrated, intelligent, and energy-efficient water purification system capable of identifying water types, optimizing purification processes, maintaining safe mineral balance, and continuously monitoring environmental water quality in real time. Existing solutions function in isolation and fail to provide end-to-end automation, adaptability, and sustainability.

Therefore, there is a critical need for a smart water purification ecosystem that combines IoT-based real-time sensing, AI/ML-driven water type identification and mineral optimization, low-power distillation, and autonomous environmental water testing. Addressing this gap is essential for improving drinking water safety, reducing health risks, minimizing resource wastage, and enabling sustainable water management, particularly in vulnerable regions such as Sri Lanka.[6]

References

[1] Google – Distilled Water

[2] Nicole Blades, Medically Reviewed by Kathleen M. Zelman, RD, LD, MPH, “Distilled Water” September 15, 2024

[3] Medically Reviewed by Shruthi N, MD, “Water Quiz: How Much Do You Know About Hydration”, October 23, 2024

[4] Google - What is distilled water used for at home?

[5] Google - distilled water vs ph7 water vs alkaline water

[6] Medically reviewed by Angelica Batingit, MD – Written by Rena Goldman, “Overview of Alkaline water, and Where to Gelt it”, Updated on November 14, 2024

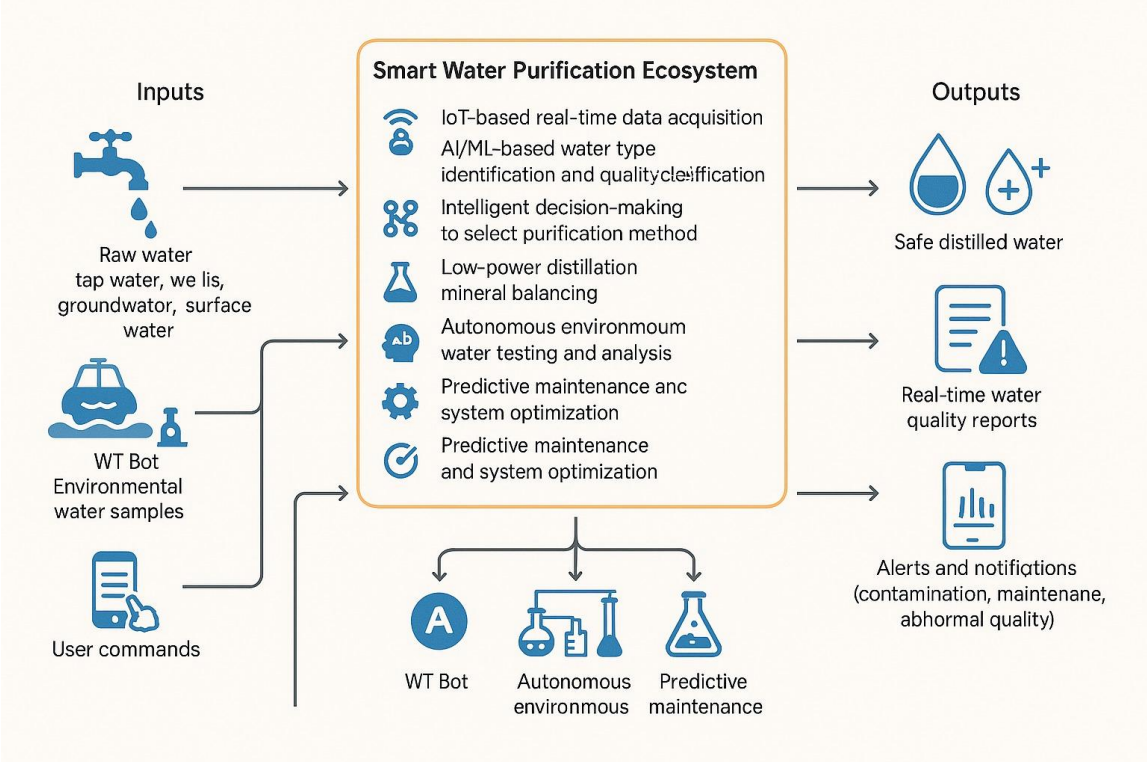
6. Brief description of the nature of the solution including a conceptual diagram (500 words max)

The proposed solution is a Smart Water Purification Ecosystem that integrates Internet of Things (IoT), Artificial Intelligence (AI), and Machine Learning (ML) technologies to deliver an intelligent, energy-efficient, and adaptive approach to drinking water purification and monitoring. Unlike conventional standalone purification systems, the proposed solution operates as an end-to-end platform capable of identifying water types, selecting suitable purification methods, enhancing water quality through controlled mineral balancing, and continuously monitoring environmental water sources in real time.

The system is designed to address critical limitations of existing purification methods, such as lack of real-time quality verification, excessive energy and water wastage, mineral imbalance, and absence of environmental surveillance. By combining real-time sensing, data-driven intelligence, automated control, and autonomous testing, the solution ensures safe, mineral-balanced, and health-appropriate water while supporting sustainable water resource management.

At the core of the system is an AI-based water type identification module, which analyzes real-time sensor data to classify incoming water (e.g., distilled, alkaline, potable, or contaminated). Based on this classification, the system intelligently activates the appropriate treatment process, including a low-power household distilled water generator, an AI-controlled alkaline mineral dosing unit, or alert mechanisms when contamination is detected. In parallel, an Autonomous Water Testing Bot (WT Bot) extends the system's capability beyond households by collecting and analyzing water samples from environmental sources such as rivers, wells, and reservoirs.

The system also includes a mobile application interface, allowing users to monitor water quality, system status, energy consumption, and alerts in real time. Predictive maintenance features powered by ML models further enhance reliability by identifying faults, scaling issues, or sensor anomalies before system failure occurs. Overall, the proposed solution provides a scalable, intelligent, and sustainable water purification framework suitable for both urban and rural environments, particularly in regions facing water quality–related health risks.


Inputs
Raw water
tap water, we lis,
groundwator, surface
water
WT Bot
Environmental
water samples
User commands
Smart Water Purification Ecosystem
IoT-based real-time data acquisition
AI/ML-based water type
identification and qualitycleiffication
Intelligent decision-making
to select purification method
Low-power distillation
mineral balancing
Autonomous environmoum
water testing and analysis
Predictive maintenance anc
system optimization
Predictive maintenance
and system optimization
WT Bot
Autonomous
environmous
Predictive
maintenance
Outputs
Safe distilled water
Real-time water
quality reports
Alerts and notifiqtions
(contamination, maintenane,
abhormal quality)

7. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

The successful execution of the Smart Water Purification Ecosystem project leverages a combination of academic resources, self-directed learning, and collaborative partnerships to acquire the necessary expertise and data.

Acquiring Specialized Domain Expertise (Water Chemistry & Public Health):

Essential knowledge in water quality parameters (pH, TDS, fluoride, pathogens), purification chemistry, and health impacts (e.g., CKDu, mineral deficiency) will be gained through multiple channels. These include:

Academic Consultation: Engaging with faculty from the Department of Biotechnology and Environmental Engineering, who possess direct expertise in Sri Lanka's water contamination challenges.

Literature Review: Conducting an in-depth review of peer-reviewed journals, Sri Lankan government reports (e.g., from the Water Supply & Drainage Board, CGHR), and international case studies on IoT for water.

Industry Engagement: Consulting with professionals from local water purification companies and NGOs (like WaterAid Sri Lanka) to understand practical constraints and real-world user needs.

Gaining Technical Knowledge & Skills (IoT, AI/ML, Robotics):

The core technical competencies required are aligned with the team's IT specialization and will be developed through:

Structured Learning: Utilizing online platforms (Coursera, Udacity) for advanced courses in Embedded Systems (Arduino/Raspberry Pi), IoT protocols (MQTT), and Machine Learning (supervised classification, regression models).

Practical Workshops & Labs: Using the university's computing and electronics laboratories for hands-on prototyping of sensor circuits, data acquisition systems, and robotic chassis control.

Supervisor & Peer Guidance: The project supervisor will provide foundational guidance, while collaborative problem-solving within the team and with senior students will address specific technical hurdles.

Data Collection Strategy:

The project requires two primary data types:

1. Water Quality Sensor Data: Initial training and validation data for the AI model will be collected from publicly available datasets (e.g., UCI Machine Learning Repository, Kaggle) and through controlled lab experiments using calibrated sensors to simulate various contaminated water samples (varying pH, turbidity, TDS). For field validation, partnerships will be sought with local authorities to collect anonymized, historical water quality data from selected regions (e.g., Anuradhapura for CKDu-prone areas).

2. System Performance Data: Data on energy consumption, purification efficiency, and mineral dosing accuracy will be generated internally during the prototype testing phase.

Ethical Clearance Consideration:

This project will require ethical clearance. While initial prototype testing will use synthetic or lab-created water samples, any subsequent field deployment of the Water Testing Bot (WT Bot) for environmental sampling or pilot testing with human users for water tasting/feedback involves ethical considerations. An application will be submitted to the Faculty/University Ethics Review Committee, addressing:

Environmental Safety: Ensuring the WT Bot does not disturb ecosystems.

Data Privacy: Anonymizing any location-specific data if published.

Informed Consent: Obtaining clear consent if community members are involved in user testing phases.

The ethical clearance process will be initiated early in the project timeline to ensure compliance.

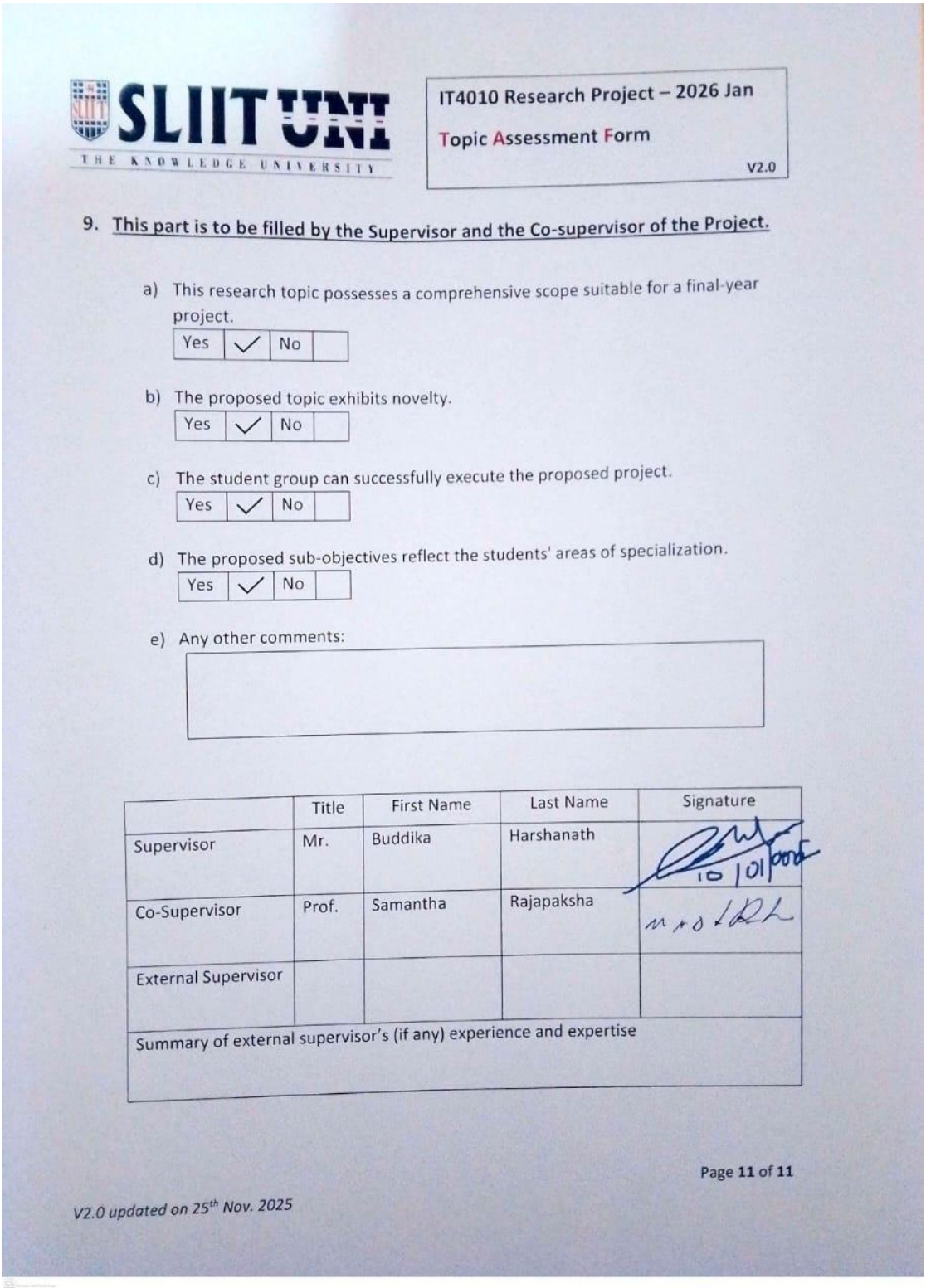
8. Objectives and Novelty

Main Objective The main objective of this research is to design and develop an intelligent, energy-efficient, and adaptive smart water purification ecosystem by integrating Internet of Things (IoT), Artificial Intelligence (AI), Machine Learning (ML), and autonomous systems to ensure the availability of safe, mineral-balanced, and high-quality drinking water. The proposed system aims to automatically identify different types of water, optimize low-power household water distillation, intelligently generate alkaline water through AI-based mineral balancing, and enable autonomous real-time environmental water quality monitoring. By leveraging real-time sensor data and intelligent decision-making, the system seeks to overcome the limitations of conventional water purification methods, reduce water and energy wastage, enhance user awareness, and support sustainable water resource management, particularly in regions facing water quality challenges such as Sri Lanka.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
W.N.N. De Silva IT22211750	AI-Based Different Types of Water Identifier	Identify key water quality parameters (pH, TDS, turbidity, conductivity, temperature). Collect and preprocess sensor data from different water sources (distilled, alkaline, groundwater, contaminated).	The novelty lies in replacing traditional manual or threshold-based water classification with an **intelligent, data-driven ML model** capable of real-time decision-making. Unlike existing systems that treat all incoming water uniformly, this approach enables **dynamic identification and context-aware**

		Design and train ML models (e.g., decision tree, SVM, or neural networks) for water type classification. Validate the model accuracy using real-world and laboratory-tested samples. Integrate the trained model with IoT sensors for real-time water type identification. Enable adaptive system responses based on identified water type (e.g., purification, mineral addition).	**processing**, improving purification efficiency and safety. The model's ability to learn from diverse datasets and adapt to regional water characteristics (e.g., Sri Lankan groundwater) further enhances its uniqueness.
W.A.S.P.D. Wijesingha IT22295460	Low-Power Household Distilled Water Generator with Mobile App Control (IoT)	Design a compact household distillation unit optimized for low power usage. Integrate temperature, energy consumption, and water level sensors. Develop IoT-based monitoring for real-time system performance.	This function introduces novelty by combining **energy-efficient distillation** with **intelligent IoT-based automation**, addressing the high-power consumption issue of conventional distillers. The use of AI for energy optimization and predictive maintenance, along with mobile app control, transforms distillation from a passive process into a **smart, user-centric, and sustainable household solution**.

		Implement AI-assisted control logic to optimize heating cycles and reduce energy waste. Develop a mobile application for remote control, monitoring, and alerts. Implement predictive maintenance features to detect faults and scaling issues early.	
D. Dodanwela IT22212122	Intelligent Alkaline Water Generation Using AI-Based Mineral Balancing (IoT & AI/ML)	Identify essential minerals (e.g., calcium, magnesium, potassium) required for safe alkaline water. Monitor real-time water quality parameters using IoT sensors. Develop AI/ML models to calculate optimal mineral dosage based on incoming water quality. Implement automated mineral dosing mechanisms.	The novelty of this module lies in its **adaptive AI-based mineral balancing mechanism**, which replaces static mineral cartridges used in conventional alkaline systems. By dynamically adjusting mineral composition in real time, the system ensures **safe alkalinity, avoids over-mineralization**, and supports personalized water quality control. This intelligent approach aligns water enhancement with both health and regulatory requirements, making it a next-generation solution.

		Ensure pH control within WHO-recommended drinking water standards. Continuously learn and adjust mineral balancing based on feedback data.	
R.M.N.A. Rathnayake IT22193940	Autonomous Water Testing Bot (WT Bot) for Real-Time Environmental Monitoring	Design an autonomous robotic platform capable of navigating different environments. Integrate water quality sensors for pH, turbidity, TDS, and microbial indicators. Enable autonomous sample collection from wells, rivers, tanks, and flood-affected areas. Implement wireless data transmission to a central monitoring system. Apply AI-based analysis for anomaly detection and contamination alerts. Support geo-tagging and time-stamped water quality reporting.	The WT Bot introduces novelty through its **mobile, autonomous, and scalable water monitoring approach**. Unlike stationary systems, it enables **on-demand and location-independent water quality assessment**, making it highly suitable for disaster response, rural monitoring, and large-scale environmental surveillance. The combination of robotics, AI analytics, and real-time data transmission significantly enhances early contamination detection and decision-making capabilities.



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9. This part is to be filled by the Supervisor and the Co-supervisor of the Project.

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes	✓	No	

b) The proposed topic exhibits novelty.

Yes	✓	No	

c) The student group can successfully execute the proposed project.

Yes	✓	No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes	✓	No	

e) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor	Mr.	Buddika	Harshanath	10/10/2025
Co-Supervisor	Prof.	Samantha	Rajapaksha	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

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